Milestone 3: Club/Ball Separation Logic

Golf Launch Monitor DSP Algorithm Development

Technical Specification Document

1 Milestone Objectives

1.1 Primary Goal

Develop robust club/ball separation algorithms that can reliably distinguish between golf club head velocity and ball velocity signatures in radar I/Q data, with particular focus on challenging wedge shots.

1.2 Core Components

- **Impact Detection**: Identify precise moment of club-ball contact using energy burst analysis
- **Timing-based Separation**: Use temporal characteristics to distinguish club vs ball signatures
- **Classification Framework**: Implement decision algorithms for club/ball assignment
- **Physics-based Validation**: Apply golf ball impact physics for consistency checking

2 Acceptance Criteria

2.1 Performance Requirements

1. **Wedge Classification**: Achieve $\geq 80\%$ correct classification accuracy on wedge shots
2. **Iron/Driver Classification**: Achieve $\geq 90\%$ correct classification accuracy on iron and driver shots
3. **Velocity Separation**: Separate club and ball velocities with < 10 mph error vs TrackMan
4. **Impact Timing**: Detect impact events within ± 2 ms of theoretical timing

2.2 Technical Validation

- Demonstrate consistent performance across velocity range: 56.5-178.9 mph
- Validate separation logic against 130+ shot dataset with TrackMan reference
- Show significant improvement over Milestone 2's basic detection
- Handle edge cases: mishits, partial contact, practice swings

3 Mathematical Framework

3.1 Impact Detection Algorithm

3.1.1 Energy Burst Detection

The total spectral energy at time t is computed as:

$$E(t) = \sum_f |STFT(f,t)|^2 \tag{1}$$

The temporal energy gradient identifies impact events:

$$\nabla E(t) = \frac{E(t+\Delta t) - E(t-\Delta t)}{2\Delta t} \tag{2}$$

Impact detection criterion:

$$\text{Impact detected} = \nabla E(t) > \alpha \cdot \sigma_{\text{noise}} \tag{3}$$

Where $\alpha = 5 - 10$ is the sensitivity factor and σ_{noise} is the background noise standard deviation.

3.1.2 Multi-Channel Impact Fusion

For 4-channel I/Q data, the fused impact energy is:

$$E_{\text{fused}}(t) = \sum_{c=1}^{4} w_c \cdot E_c(t) \tag{4}$$

With channel weights $w_c = [0.25, 0.25, 0.25, 0.25]$ for equal contribution.

3.2 Velocity Signature Analysis

3.2.1 Continuity Metric

The velocity continuity metric distinguishes smooth club motion from impulsive ball motion:

$$C = \frac{1}{N-1} \sum_{i=1}^{N-1} |v(t_{i+1}) - v(t_i)| \tag{5}$$

Classification rule:

$$C < \tau_{\text{smooth}} \Rightarrow \text{Club signature} \tag{6}$$
$$C > \tau_{\text{impulsive}} \Rightarrow \text{Ball signature} \tag{7}$$

3.2.2 Peak Velocity Ratio Analysis

The velocity ratio provides club type classification:

$$R_v = \frac{\max(v_{\text{ball}})}{\max(v_{\text{club}})} \tag{8}$$

Expected ranges:

$$\text{Wedge shots:} \quad 1.2 \le R_v \le 1.8 \tag{9}$$
$$\text{Iron shots:} \quad 1.4 \le R_v \le 2.2 \tag{10}$$
$$\text{Driver shots:} \quad 1.6 \le R_v \le 2.5 \tag{11}$$

3.3 Spectral Characteristics Analysis

3.3.1 Spectral Width Computation

The spectral width distinguishes extended club contact from point ball target:

$$W_s = \sqrt{\frac{\sum_f f^2|X(f)|^2}{\sum_f |X(f)|^2} - \left(\frac{\sum_f f|X(f)|^2}{\sum_f |X(f)|^2}\right)^2} \tag{12}$$

Classification criterion:

$$W_s > \tau_{\text{broad}} \Rightarrow \text{Club signature (extended contact)} \tag{13}$$
$$W_s < \tau_{\text{narrow}} \Rightarrow \text{Ball signature (point target)} \tag{14}$$

3.4 Cross-Correlation Analysis

3.4.1 Temporal Correlation

Cross-correlation between club and ball velocity tracks:

$$R_{cb}(\tau) = \frac{\sum_t v_{\text{club}}(t) \cdot v_{\text{ball}}(t+\tau)}{\sqrt{\sum_t v_{\text{club}}^2(t) \cdot \sum_t v_{\text{ball}}^2(t)}} \tag{15}$$

The time delay corresponds to impact physics:

$$\tau_{\text{delay}} = \arg\max_{\tau} R_{cb}(\tau) \tag{16}$$

Expected delay: $1 - 3$ ms for golf ball impact duration.

3.5 Feature Vector Construction

For each detected track, the feature vector is:

$$\mathbf{f} = \begin{bmatrix} v_{\max} \\ t_{\text{rise}} \\ W_s \\ t_{\text{impact}} \\ C \\ T_{\text{duration}} \\ \tau_{\text{delay}} \end{bmatrix} \tag{17}$$

Where:

$$v_{\max} = \text{peak velocity} \tag{18}$$
$$t_{\text{rise}} = \text{time to reach peak velocity} \tag{19}$$
$$W_s = \text{spectral width} \tag{20}$$
$$t_{\text{impact}} = \text{relative timing to impact event} \tag{21}$$
$$C = \text{continuity metric} \tag{22}$$
$$T_{\text{duration}} = \text{track temporal extent} \tag{23}$$
$$\tau_{\text{delay}} = \text{cross-correlation delay} \tag{24}$$

3.6 Classification Algorithms

3.6.1 Rule-Based Decision Tree

$$\text{if } (t_{\text{rise}} < 2 \text{ ms AND } v_{\max} > v_{\text{club\_threshold}}) \Rightarrow \text{Ball} \tag{25}$$

$$\text{else if } (C < \tau_{\text{smooth}}) \Rightarrow \text{Club} \tag{26}$$

$$\text{else} \Rightarrow \text{Use ML classifier} \tag{27}$$

3.6.2 Support Vector Machine

For complex cases, apply SVM with RBF kernel:

$$K(\mathbf{f}_i, \mathbf{f}_j) = \exp\left(-\gamma ||\mathbf{f}_i - \mathbf{f}_j||^2\right) \tag{28}$$

Decision function:

$$f(\mathbf{x}) = \sum_{i=1}^{N} \alpha_i y_i K(\mathbf{f}_i, \mathbf{x}) + b \tag{29}$$

4 Algorithm Parameters

4.1 Impact Detection Parameters

Parameter	Value	Description
Energy gradient threshold (α)	7.0	Impact sensitivity factor
Analysis window	$\pm$10 ms	Impact timing window
Minimum impact duration	0.3 ms	Physical impact constraint
Channel fusion weights	[0.25, 0.25, 0.25, 0.25]	Equal channel contribution

4.2 Velocity Analysis Parameters

Parameter	Value	Description
Velocity window	$\pm$50 ms	Analysis window around impact
Continuity threshold (smooth)	2.0 mph/ms	Club motion threshold
Continuity threshold (impulsive)	15.0 mph/ms	Ball motion threshold
Rise time threshold	2.0 ms	Ball acceleration criterion

4.3 Spectral Analysis Parameters

Parameter	Value	Description
FFT size	1024	Spectral resolution
Zero-padding factor	2	Enhanced resolution
Spectral width threshold (broad)	50 Hz	Club signature criterion
Spectral width threshold (narrow)	20 Hz	Ball signature criterion

Parameter	Value	Description
Decision tree max depth	5	Complexity limit
SVM RBF gamma (γ)	0.1	Kernel parameter
SVM regularization (C)	1.0	Penalty parameter
Random Forest trees	100	Ensemble size
Confidence threshold	0.7	High-confidence classification
Cross-correlation lag range	$\pm$10 ms	Temporal search window

4.4 Classification Parameters

5 Dataset Integration

5.1 Input Data Structure

Building upon the established dataset schema:

- **Binary Files**: `LOG[YYYYMMDD][HHMMSS]_[SEQUENCE].bin`
- **Metadata**: JSON files with TrackMan reference speeds
- **Milestone 2 Results**: Multi-target tracking outputs
- **Velocity Range**: 56.5-178.9 mph (ball), 45.7-120.4 mph (club)

5.2 Processing Pipeline Integration

1. Load Milestone 2 tracking results and I/Q data
2. Apply impact detection algorithms on spectral energy
3. Extract velocity and spectral features from each track
4. Compute cross-correlation and temporal characteristics
5. Apply rule-based and ML classification algorithms
6. Validate separation results against TrackMan reference

6 Expected Visualizations

6.1 Impact Detection Analysis

- **Energy Burst Detection**: Time-frequency spectrograms with detected impact events
- **Multi-Channel Fusion**: Energy contributions from each I/Q channel
- **Impact Timing Validation**: Detected vs. theoretical impact timing scatter plots
- **Gradient Analysis**: Temporal energy gradients showing impact signatures

6.2 Feature Space Analysis

- **Feature Distributions**: Histograms of club vs. ball feature values
- **Feature Correlation Matrix**: Cross-correlations between extracted features
- **Principal Component Analysis**: 2D/3D visualization of feature space
- **Feature Importance**: Rankings for different club types and classification methods

6.3 Classification Performance

- **Confusion Matrices**: Classification accuracy by club type (wedge/iron/driver)
- **ROC Curves**: Receiver operating characteristic analysis for binary classification
- **Precision-Recall Curves**: Performance trade-offs for different thresholds
- **Classification Confidence**: Distribution of classification certainty scores

6.4 Velocity Separation Results

- **Separation Accuracy**: Club/ball velocity estimates vs. TrackMan scatter plots
- **Error Analysis**: RMS error distributions by shot type and velocity range
- **Velocity Trajectories**: Time-series plots showing separated club/ball tracks
- **Physics Validation**: Velocity ratios and energy transfer consistency checks

6.5 Performance Comparison

- **Milestone Progression**: Improvement over Milestone 2 basic detection
- **Algorithm Comparison**: Rule-based vs. ML classification performance
- **Club Type Analysis**: Performance breakdown by wedge/iron/driver categories
- **Failure Mode Analysis**: Systematic study of misclassification patterns

7 Success Metrics

7.1 Primary Classification Targets

- **Wedge Accuracy**: $\geq 80\%$ correct club/ball classification
- **Iron Accuracy**: $\geq 90\%$ correct club/ball classification
- **Driver Accuracy**: $\geq 90\%$ correct club/ball classification
- **Overall Dataset**: $\geq 85\%$ correct classification across all shots

7.2 Velocity Separation Accuracy

- **Club Velocity Error**: < 10 mph RMS vs. TrackMan reference
- **Ball Velocity Error**: < 10 mph RMS vs. TrackMan reference
- **Separation Consistency**: $< 15\%$ standard deviation across shots
- **Impact Timing**: ± 2 ms accuracy for impact event detection

7.3 Robustness Requirements

- Consistent performance across full velocity range (56.5-178.9 mph)
- Stable classification under varying signal-to-noise conditions
- Graceful handling of edge cases (mishits, partial contact, multiple balls)
- Computational efficiency suitable for real-time embedded implementation

8 Deliverables

- Complete MATLAB source code with full documentation
- Trained machine learning models and classification thresholds
- Performance analysis report with comprehensive visualizations
- Club/ball separation validation results against TrackMan dataset
- Algorithm parameter sensitivity analysis and optimization recommendations
- Implementation guidance for Milestone 4 wedge-specific enhancements